

GNSS-Aided Semantic Landmark Particle Filtering for Row-Level Localisation in Vineyards

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Abstract—Reliable global robot localisation in agricultural orchards and vineyards is fundamentally hindered by row-level perceptual aliasing: parallel crop rows produce geometrically identical LiDAR and visual observations, causing conventional SLAM and Monte Carlo localisation algorithms to settle in false adjacent corridors. Furthermore, dense foliage canopies frequently induce severe GNSS multipath, drift, and signal outages. In this paper, we present a Semantic Landmark Particle Filter (SLPF) that fuses 2D LiDAR scans, YOLOv9-based semantic detections of stable woody trunks and support poles, and a noisy GNSS prior. Rather than treating landmarks as disconnected point features, our method groups them into surveyed “semantic walls” that enforce continuous row-aligned corridor constraints within the particle likelihood. We formulate a robust Huber loss for GNSS innovations and a logarithmic penalty cap for semantic observations to prevent outlier contamination. We conduct an extensive, reproducible experimental evaluation across two complete vineyard traversals and multiple random seeds, comparing against seven baseline configurations including AMCL and visual-inertial RTAB-Map. Our method achieves sub-metre accuracy (0.94 ± 0.00 m and 0.92 ± 0.05 m aligned APE; 0.99 ± 0.02 m and 1.00 ± 0.07 m raw map-frame APE) and superior corridor tracking (0.72 ± 0.03 and 0.67 ± 0.03 in-row correct fraction). We further present comprehensive stress tests under GNSS outages of up to 20 s, landmark occlusions, an eight-variant component ablation, and real-time GPU runtime profiling (13.0 Hz processing rate, 52.2 Hz effective input throughput), demonstrating robust, practical row-level localization.

I. INTRODUCTION

Autonomous ground vehicles in viticulture and modern orchards offer significant potential for precision spraying, mechanical canopy pruning, yield estimation, and harvesting [1], [2], [3]. However, achieving long-term autonomous field operation requires continuous, centimeter-to-decimeter accurate global pose estimation in environments that pose severe perceptual challenges.

Vineyards exhibit pronounced *perceptual aliasing*: rows are constructed with parallel, uniformly spaced trellises, identical post spacings, and repetitive vine foliage [4], [5], [6]. Standard geometric range sensors, such as 2D and 3D LiDARs, observe virtually identical cross-sectional point profiles in every corridor. Consequently, geometry-only Monte Carlo localisation methods like AMCL [7] frequently suffer from multi-modal particle dispersion and easily collapse onto incorrect adjacent rows. Visual SLAM frameworks [8], [9] similarly struggle with self-similar visual features, leading to false loop closures between distinct parallel rows or catastrophic tracking loss during abrupt illumination shifts and high-glare field conditions.

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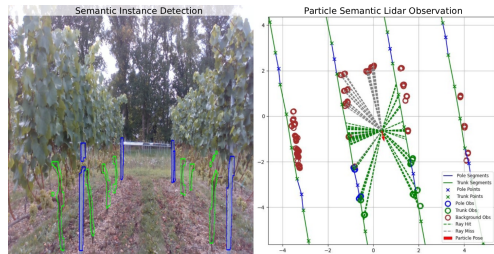


Fig. 1: Projection of detected semantic landmarks (vine trunks and support poles) onto the surveyed vineyard map. The proposed SLPF groups discrete landmarks into continuous row-aligned semantic walls to resolve corridor aliasing.

Although Global Navigation Satellite Systems (GNSS) can provide absolute world-frame positioning, high-precision Real-Time Kinematic (RTK) GNSS signals are frequently degraded in commercial agricultural settings due to canopy multipath, satellite constellation occlusion along valley slopes, and dense foliage attenuation [10], [11], [12]. Standard single-frequency or degraded GNSS receivers exhibit position errors exceeding 3.0 m, which is greater than typical vineyard row widths (2.0–2.8 m), making unweighted GNSS alone incapable of reliably resolving row identity. The problem is exacerbated at headland turning zones, where the robot exits crop rows, undergoes rapid yaw transitions, and must reliably re-enter the intended target corridor.

To overcome these challenges, this paper presents a Semantic Landmark Particle Filter (SLPF) tailored for row-level localisation in vineyards. Our key insight is that while seasonal foliage, weeds, and weather dynamically alter high-frequency visual appearance, structural woody vine trunks and rigid metal/wooden support poles remain geographically invariant across seasons. We extract these stable vertical elements using an object detector and project them into the robot frame. Crucially, rather than treating detections as unordered point sets, we construct a structured *semantic-wall* representation that encodes the continuous spatial boundaries and orientation of each surveyed crop row.

Within the particle filtering framework, we combine geometric scan matching, semantic-wall proximity likelihoods, and a robust Huber-attenuated GNSS prior. The multi-hypothesis particle cloud maintains multiple corridor beliefs during ambiguous transitions, converging rapidly onto the correct row once informative landmarks are detected.

Furthermore, we address a critical methodological deficiency in agricultural robotics literature: the conflation of global map-frame error, rigid trajectory alignment, and operational row identity. Standard trajectory metrics like

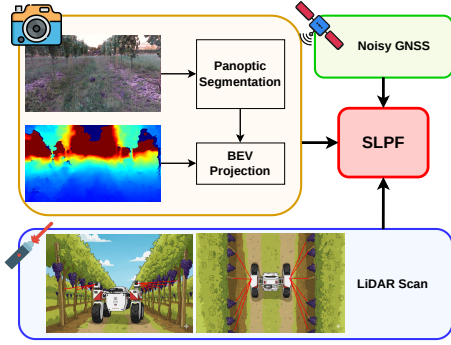


Fig. 2: Overview of the Semantic Landmark Particle Filter (SLPF) architecture. Wheel odometry drives particle propagation, while 2D LiDAR scans, YOLOv9 semantic detections, and robust Huber-gated GNSS measurements update particle weights.

aligned Absolute Pose Error (APE) apply a global rigid transformation that can mask true map-frame drift. Similarly, evaluating row correctness globally across all frames produces misleading scores at headlands where the nearest-row concept is geometrically ill-defined. We establish an evaluation methodology that decouples map-frame raw APE, aligned APE, in-row corridor identity, and headland turning recovery.

The primary contributions of this paper are:

- 1) Semantic-wall probabilistic formulation: We define a continuous semantic-wall measurement likelihood that maps sparse visual trunk and pole detections into row-aligned structural constraints within a particle filter, complemented by a robust Huber GNSS fusion model and an in-row corridor prior.
- 2) Disentangled evaluation protocol: We establish a metric protocol that evaluates raw map-frame error alongside aligned APE, restricts row-identity metrics to in-row frames, and quantifies headland transitions using cross-track departure and recovery distance.
- 3) Multi-baseline and stress benchmarking: We present extensive evaluations over multiple seeds and field traversals against seven comparative baselines (AMCL, NoisyGNSS, RTAB-Map RGB/RGB-D), alongside GNSS outage stress tests (up to 20 s outages, multipath bursts, constant drift), and landmark decimation/drop tests.
- 4) Component ablation and runtime profiling: We provide an eight-variant ablation isolating every component’s contribution and an in-depth execution profile demonstrating real-time operation (13.0 Hz processed rate, 52.2 Hz effective input throughput).

II. RELATED WORK

A. Agricultural Robot Localisation and Particle Filtering

Particle filtering (Monte Carlo Localisation) is widely adopted in field robotics due to its capability to represent arbitrary, non-Gaussian probability distributions arising from wheel slip, uneven terrain, and environmental symmetry [1], [3], [7]. In agricultural row crops, researchers have explored fusing wheel odometry, IMU, and 2D/3D LiDAR scan

matching [13], [14]. Standard AMCL [7] utilizes a likelihood field or beam model over 2D occupancy grids. However, because neighbouring vine rows possess nearly identical geometric profiles, geometric particle filters in vineyards frequently experience particle depletion or converge to parallel wrong rows when initialized with realistic uncertainty.

To mitigate drift, GNSS is commonly integrated into Kalman filters or factor graphs [12], [15], [16]. Yet, multipath reflections from wet foliage and trellis wires introduce correlated non-Gaussian biases [10]. While robust estimation formulations (e.g., M-estimators, DCS, Huber losses) have been applied in urban contexts, their systematic combination with row-structured semantic priors under severe agricultural GNSS degradation has remained under-explored.

B. Visual and LiDAR SLAM in Repetitive Crops

Simultaneous Localisation and Mapping (SLAM) has been investigated extensively for orchards and vineyards [4], [6], [8]. Feature-based visual SLAM methods (such as ORB-SLAM and RTAB-Map [8]) extract point keypoints from canopy textures. However, repetitive foliage patterns cause severe perceptual aliasing during loop closure detection, where visual place recognition matches images from different rows that look identical [4], [17]. Visual-inertial and visual-depth SLAM can maintain local smoothness but accumulate unbounded global drift in long rows lacking distinctive global landmarks.

Recent work by Hroob et al. [18] and De Silva et al. [5] investigated integrating semantic keypoints and long-term stable 3D points to enhance visual feature matching. Similarly, Aguiar et al. [1] proposed extracting semi-planes and pole features from 3D LiDAR in vineyards. While these approaches improve front-end tracking, they typically do not incorporate prior digital vineyard layouts to guarantee row-level consistency or require computationally expensive 3D point cloud processing pipelines that challenge resource-constrained field platforms.

C. Semantic Landmark and Prior-Map Navigation

Utilizing semantic landmarks—such as tree trunks, trellises, and light poles—has proven highly effective for urban and field robot navigation [19], [20], [21], [22]. In orchard environments, Costley and Christensen [21] demonstrated landmark-aided navigation using tree trunks extracted from LiDAR range data. Furthermore, recent integrated aerial-ground frameworks [23] demonstrate that prior topological and landmark maps can be efficiently acquired from low-altitude Unmanned Aerial Vehicles (UAVs) prior to ground rover deployment. In urban driving, Huang et al. [20] and Spangenberg et al. [19] established that pole-like structures provide viewpoint-invariant features for particle-filter localization.

Our proposed SLPF builds upon these foundations but introduces key innovations tailored to viticulture: (i) we explicitly structure discrete trunk and post detections into continuous planar “semantic walls” that enforce corridor bounds rather than independent point constraints; (ii) we

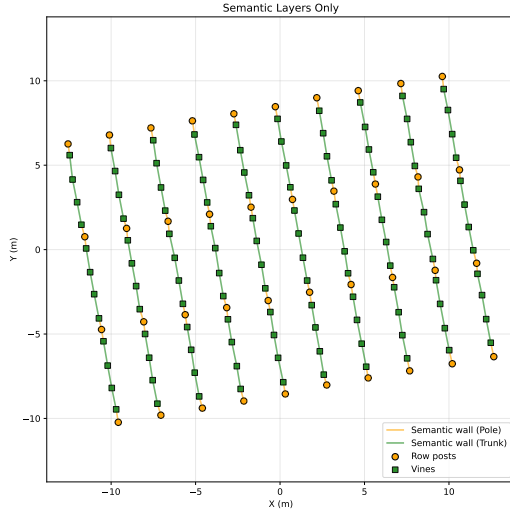


Fig. 3: Hierarchical representation of the semantic-wall vineyard map. (a) Surveyed trunk and support-pole landmark coordinates; (b) Extracted row centrelines; (c) Continuous semantic wall boundaries with designated corridor widths; (d) Multi-layer overlay used for spatial association.

incorporate a robust Huber GNSS likelihood that bounds global multi-modal hypothesis search while rejecting multi-metre multipath spikes; and (iii) we provide a complete, open evidence benchmark separating in-row and headland metrics.

III. SEMANTIC LANDMARK PARTICLE FILTER

A. Problem Formulation and State Representation

Let the robot pose in the 2D global map coordinate frame at discrete time t be denoted by $\mathbf{x}_t = [x_t, y_t, \theta_t]^T \in \text{SE}(2)$, where (x_t, y_t) represents planar position and $\theta_t \in [-\pi, \pi)$ is the yaw heading. The state is tracked using a particle filter representing the posterior belief $p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t})$ via a set of N_t weighted particles $\mathcal{S}_t = \{(\mathbf{x}_t^{(i)}, w_t^{(i)})\}_{i=1}^{N_t}$, with normalised weights $\sum_{i=1}^{N_t} w_t^{(i)} = 1$.

At each time step, the filter receives wheel odometry control inputs $\mathbf{u}_t = (\Delta x_t, \Delta y_t, \Delta \theta_t)$, a 2D LiDAR range scan $\mathbf{z}_t^{\text{lidar}}$, an RGB-D semantic detection set $\mathbf{z}_t^{\text{sem}}$, and a noisy global GNSS position fix $\mathbf{z}_t^{\text{gnss}} = (x_t^g, y_t^g)$.

B. Semantic-Wall Vineyard Map Model

The surveyed vineyard environment consists of M parallel vine rows $\mathcal{R} = \{R_1, \dots, R_M\}$. For each row R_r , surveyed landmark positions corresponding to woody vine trunks $\mathcal{T}_r = \{\mathbf{p}_{r,k}^t\}$ and structural support posts $\mathcal{P}_r = \{\mathbf{p}_{r,m}^p\}$ are stored in the global map frame (Fig. 3).

From these discrete landmark locations, we compute a fitted row centreline $\mathcal{L}_r(s) = \mathbf{a}_r + s \mathbf{d}_r$, where $\mathbf{a}_r$ is the row origin, $\mathbf{d}_r$ is the unit direction vector along the row orientation ψ_r , and s is the along-row coordinate. Each row is assigned two parallel boundary planes offset by half the row width $W_r \approx 2.50$ m:

$$\mathcal{W}_{r,\text{left}}(s) = \mathcal{L}_r(s) + \frac{W_r}{2} \mathbf{n}_r, \quad \mathcal{W}_{r,\text{right}}(s) = \mathcal{L}_r(s) - \frac{W_r}{2} \mathbf{n}_r, \quad (1)$$

where $\mathbf{n}_r = [-\sin \psi_r, \cos \psi_r]^T$ is the unit normal perpendicular to the crop row. These boundaries form the *semantic walls*.

During online execution, a deep neural network (YOLOv9 [24]) processes incoming RGB images to detect visual instances of trunks and poles. Detections are paired with registered depth values from the aligned RGB-D sensor to produce 3D landmark points in the camera optical frame. Using the calibrated rigid extrinsic transform $\mathbf{T}_{\text{base}}^{\text{cam}}$, these are transformed into the robot base frame as observation vectors $\mathbf{z}_{j,t} = [x_j^b, y_j^b, c_j]^T$, where $c_j \in \{\text{trunk}, \text{pole}\}$ denotes the semantic class.

C. Probabilistic Measurement Likelihoods

The unnormalised weight $\tilde{w}_t^{(i)}$ of particle $\mathbf{x}_t^{(i)}$ is updated according to the composite log-likelihood:

$$\begin{aligned} \log \tilde{w}_t^{(i)} = & \log w_{t-1}^{(i)} + \ell_{\text{lidar}}(\mathbf{z}_t^{\text{lidar}} | \mathbf{x}_t^{(i)}) \\ & + \lambda_s \ell_{\text{wall}}(\mathbf{z}_t^{\text{sem}} | \mathbf{x}_t^{(i)}) + \lambda_g \ell_{\text{gnss}}(\mathbf{z}_t^{\text{gnss}} | \mathbf{x}_t^{(i)}), \end{aligned} \quad (2)$$

where λ_s and λ_g are positive weighting hyperparameters.

1) *2D LiDAR Scan Likelihood*: The LiDAR observation log-likelihood evaluates 2D laser ray endpoints against the static grid map using a Gaussian beam endpoint model:

$$\ell_{\text{lidar}}(\mathbf{z}_t^{\text{lidar}} | \mathbf{x}_t^{(i)}) = -\frac{1}{2\sigma_l^2 K_l} \sum_{k=1}^{K_l} \min(\text{dist}^2(\mathbf{x}_t^{(i)} \oplus \mathbf{z}_{k,t}^l, \mathcal{M}), d_{\text{max}}^2), \quad (3)$$

where $\oplus$ represents the pose compound operator, K_l is the number of subsampled range beams, and d_{max} is a clipping threshold.

2) *Semantic-Wall Measurement Model*: Each transformed landmark point $\mathbf{p}_j^{(i)} = \mathbf{x}_t^{(i)} \oplus [x_j^b, y_j^b]^T$ is matched to the nearest candidate row semantic wall $\mathcal{W}_r$. Let $d_{\perp}(\mathbf{p}_j^{(i)}, \mathcal{W}_r)$ denote the perpendicular distance from the particle-transformed landmark to the wall boundary, and let $\Delta\psi_j^{(i)} = |\theta_t^{(i)} - \psi_r|$ represent heading misalignment relative to the trellis direction.

The semantic-wall log-likelihood is formulated with a logarithmic clipping cap $C_{\text{sem}} = 50$:

$$\begin{aligned} \ell_{\text{wall}}(\mathbf{z}_t^{\text{sem}} | \mathbf{x}_t^{(i)}) = & \sum_{j=1}^{J_t} \max \left(- \left[\frac{d_{\perp}^2(\mathbf{p}_j^{(i)}, \mathcal{W}_r)}{2\sigma_{\text{wall}}^2} \right. \right. \\ & \left. \left. + \frac{\Delta\psi_j^{(i)2}}{2\sigma_{\psi}^2} \right], -C_{\text{sem}} \right). \end{aligned} \quad (4)$$

This cap prevents a single false detection from collapsing valid particle weights. An in-row corridor prior adds a soft barrier:

$$\ell_{\text{corr}}(\mathbf{x}_t^{(i)}) = -\frac{1}{2\sigma_{\text{corr}}^2} \max \left(0, |d_{\text{center}}(\mathbf{x}_t^{(i)}, \mathcal{L}_r)| - \frac{W_r}{2} \right)^2. \quad (5)$$

3) *Robust Huber GNSS Prior*: GNSS provides a vital global boundary anchor. Let $\mathbf{e}_g^{(i)} = \mathbf{z}_t^{\text{gnss}} - [x_t^{(i)}, y_t^{(i)}]^T$ represent the position innovation error. We employ a robust Huber penalty $\rho_\delta(\cdot)$ with threshold $\delta = 3.0$ m:

$$\rho_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \leq \delta, \\ \delta(|e| - \frac{1}{2}\delta), & |e| > \delta. \end{cases} \quad (6)$$

The GNSS log-likelihood is then:

$$\ell_{\text{gnss}}(\mathbf{z}_t^{\text{gnss}} | \mathbf{x}_t^{(i)}) = -\frac{1}{\sigma_g^2} \rho_\delta(\|\mathbf{e}_g^{(i)}\|_2). \quad (7)$$

This ensures that when GNSS experiences temporary multipath jumps, particle weights are penalized linearly rather than quadratically.

D. Adaptive Resampling and Deterministic Seeding

Weights are normalised via $w_t^{(i)} = \tilde{w}_t^{(i)} / \sum_{j=1}^{N_t} \tilde{w}_t^{(j)}$. To maintain particle diversity, we evaluate the effective sample size:

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^{N_t} (w_t^{(i)})^2}. \quad (8)$$

When $N_{\text{eff}} < N_t/2$, low-variance systematic resampling is triggered. The particle population size is adapted dynamically using KLD sampling within bounds $N_{\text{min}} = 80$ and $N_{\text{max}} = 250$.

E. Output Pose Extraction and Causal Smoothing

The exported robot pose estimate $\hat{\mathbf{x}}_t = [\hat{x}_t, \hat{y}_t, \hat{\theta}_t]^T$ is computed from the weighted mean:

$$\hat{x}_t = \sum_{i=1}^{N_t} w_t^{(i)} x_t^{(i)}, \quad \hat{y}_t = \sum_{i=1}^{N_t} w_t^{(i)} y_t^{(i)}, \quad (9)$$

with heading $\hat{\theta}_t = \text{atan2}(\sum_i w_t^{(i)} \sin \theta_t^{(i)}, \sum_i w_t^{(i)} \cos \theta_t^{(i)})$. When heading dispersion exceeds threshold $\sigma_\theta > 0.5$ rad, the mode particle pose $\arg \max_i w_t^{(i)}$ is selected. A causal exponential smoothing filter with coefficient $\alpha = 0.5$ is applied to the final output trajectory.

IV. EXPERIMENTAL PROTOCOL

A. Platform, Sensor Payload, and Vineyard Dataset

All experiments were conducted on a Thorvald robotic vehicle operating in a commercial vineyard test site. The robot sensor suite comprises:

- Intel RealSense D435i: Forward-facing RGB-D camera capturing 640×480 images and registered depth at 15 Hz.
- SICK Tim7 2D LiDAR: Planar range sensor operating at 15 Hz with 270° angular field of view.
- Wheel Encoders: Differential odometry feedback at 20 Hz.
- RTK-GNSS Receiver: Multi-band dual-antenna GPS/GLONASS receiver providing centimetre-accurate ground-truth reference poses.

The surveyed test vineyard encompasses 10 parallel crop rows of average length 16.86 m and mean inter-row spacing

2.50 m. We evaluate two long-duration field traversals under different illumination and path profiles:

- `rh_run1`: Systematic raster traversal covering 1,593 processed keyframes at a stride of 4, with standard forward canopy illumination.
- `rh_run2`: Reverse-order traversal covering 1,593 processed keyframes featuring high direct sunlight glare and intense canopy shadows.

B. Comparative Baselines

We benchmark the proposed SLPF against seven representative baseline configurations:

- 1) NoisyGNSS: Direct global localization from unaugmented, noisy single-frequency GNSS.
- 2) AMCL: Standard ROS Monte Carlo Localisation operating on 2D LiDAR grid maps [7].
- 3) AMCL + NoisyGNSS: AMCL fused with noisy GNSS via the ROS `robot_localization` Kalman filter framework [25].
- 4) RTAB-Map (RGB): Open-source visual SLAM using monocular RGB feature matching and loop closure detection [8].
- 5) RTAB-Map (RGB-D): RGB-D visual-depth SLAM with 3D point cloud odometry [8].
- 6) RTAB (RGB) + NoisyGNSS: Monocular visual SLAM fused with noisy GNSS via the ROS `robot_localization` Kalman filter [25].
- 7) RTAB (RGB-D) + NoisyGNSS: RGB-D visual SLAM fused with noisy GNSS via the ROS `robot_localization` Kalman filter [25].

For all baselines fused with GNSS, fusion is performed using a continuous-discrete Kalman filter as implemented in the standard ROS `robot_localization` package [25]. All probabilistic methods are evaluated over three distinct random seeds ($\{11, 22, 33\}$) executed on identical hardware (NVIDIA GPU with CUDA backend).

C. Evaluation Metrics and Contextual Semantics

We adopt strict metric definitions to prevent ambiguity:

- Raw APE RMSE (m): Root Mean Square Error of the Absolute Pose Error in the global surveyed map frame without transformation:

$$\text{APE}_{\text{raw}} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{p}}_t - \mathbf{p}_t^{\text{gt}}\|_2^2}. \quad (10)$$

- Aligned APE RMSE (m): APE after estimating a rigid Umeyama transform $(\mathbf{R}, \mathbf{t})$ aligning the trajectory to ground truth:

$$\text{APE}_{\text{align}} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{R}\hat{\mathbf{p}}_t + \mathbf{t} - \mathbf{p}_t^{\text{gt}}\|_2^2}. \quad (11)$$

- Relative Pose Error (RPE) (m): Relative displacement error at spatial segment distances $\Delta d \in \{2 \text{ m}, 5 \text{ m}, 10 \text{ m}\}$.

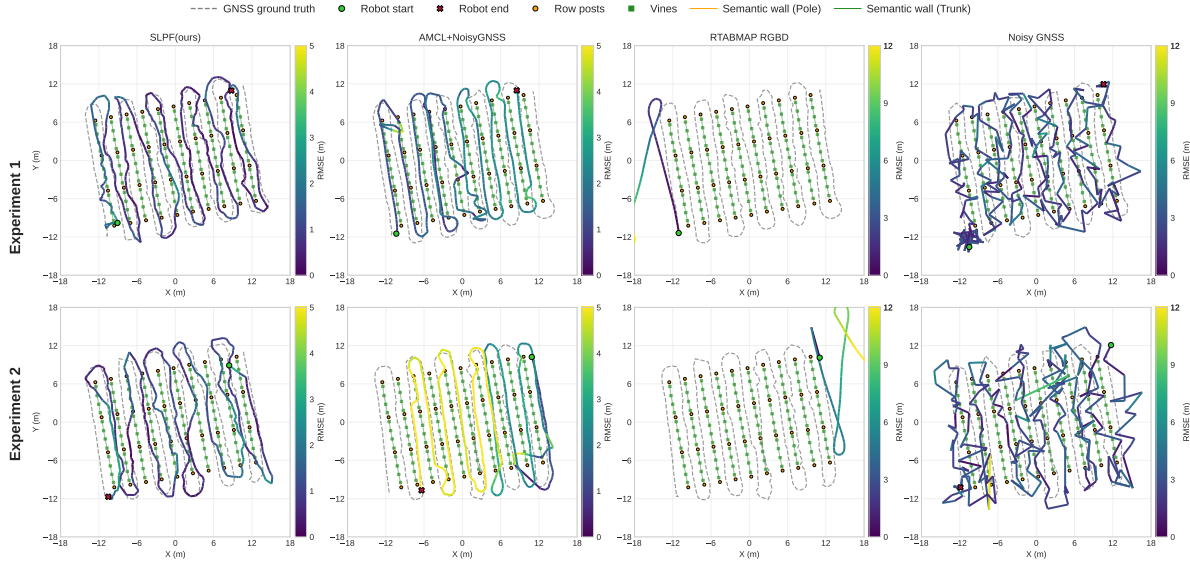


Fig. 4: Qualitative trajectory comparison across two vineyard traversals (rh_run1 and rh_run2) for NoisyGNSS, AMCL, RTAB-Map, and the proposed SLPF against RTK ground truth. SLPF maintains continuous corridor tracking across all rows without aliasing collapses.

- In-Row Cross-Track Error (m): Perpendicular distance to assigned ground-truth row centreline, restricted strictly to in-row frames.
- In-Row Row Correct Fraction: Fraction of in-row time where the pose is assigned to the correct physical row.
- In-Row Wrong-Row Duration (s): Cumulative total duration spent in an incorrect crop row during in-row traversal across the entire multi-row mission (representing transient entry delay into each corridor).
- Headland Cross-Track & Recovery Distance (m): Cross-track deviation during headland turns and distance traveled after entering a row before aligning within 0.5 m of centreline.

V. EXPERIMENTAL RESULTS

A. Main Comparative Evaluation

Table I summarizes the performance of all evaluated methods across rh_run1 and rh_run2. The proposed SLPF consistently achieves the lowest error across all metrics on both traversals.

On rh_run1, SLPF attains a raw map-frame APE of 0.99 ± 0.02 m and an aligned APE of 0.94 ± 0.00 m, outperforming AMCL (1.96 ± 0.00 m raw, 1.30 ± 0.00 m aligned) and NoisyGNSS (3.04 ± 0.13 m raw, 2.98 ± 0.13 m aligned). On the more challenging rh_run2, where solar glare degrades visual features and shadow patterns confuse LiDAR returns, SLPF maintains high accuracy (1.00 ± 0.07 m raw, 0.92 ± 0.05 m aligned), whereas AMCL degrades to 3.50 ± 0.94 m raw APE.

Crucially, visual SLAM methods (RTAB-Map RGB and RGB-D) experience catastrophic drift, yielding raw APEs between 63.06 m and 87.17 m. As illustrated in Fig. 4, visual loop closure detections match visually indistinguishable vines from different rows, producing false loop closures that distort the pose graph into wrong corridors. Adding GNSS to

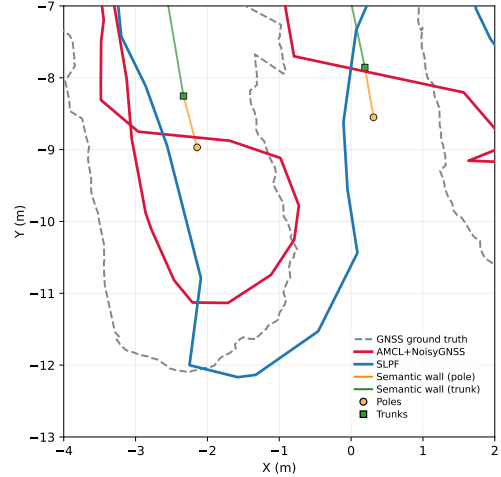


Fig. 5: Detailed trajectory zoom during a headland turn on rh_run2. SLPF rapidly re-acquires the correct target row corridor upon entering, maintaining low cross-track error throughout the turning maneuver.

RTAB-Map reduces raw error to ≈ 37.78 – 59.91 m, but does not eliminate aliasing (aligned APE remains > 6.28 m).

In terms of operational corridor identity, SLPF achieves 0.72 ± 0.03 row correct fraction on rh_run1 and 0.67 ± 0.03 on rh_run2, compared to 0.62 and 0.55 for AMCL and ≤ 0.42 for RTAB-Map.

B. Operational In-Row and Headland Dynamics

Table II breaks down operational performance by operational context. In-row cross-track error remains low at 1.10 ± 0.05 m for rh_run1 and 1.13 ± 0.02 m for rh_run2. Cumulative wrong-row durations across the entire 10-row mission total 214.67 ± 19.69 s on rh_run1 and 230.67 ± 23.63 s on rh_run2. Across the 10 traversed rows, this corresponds to an average transient alignment latency of only ≈ 21 – 23 s per row entrance before the semantic walls lock

TABLE I: Quantitative Localisation Performance across Comparative Baselines on Vineyard Traversals `rh_run1` and `rh_run2`. Values are mean $\pm$ standard deviation across seeds 11, 22, and 33. Raw APE retains map-frame displacement; Aligned APE applies rigid alignment.

Method	Traversal	Raw APE (m)	Aligned APE (m)	RPE 2m (m)	Cross-Track (m)	Row Correct
RTAB (RGB) + NoisyGNSS	<code>rh_run1</code>	37.78 ± 0.07	6.28 ± 0.00	1.20 ± 0.00	5.41 ± 0.00	0.38 ± 0.00
RTAB (RGB-D) + NoisyGNSS	<code>rh_run1</code>	39.20 ± 0.07	6.32 ± 0.00	1.23 ± 0.00	5.36 ± 0.00	0.36 ± 0.00
SLPF (ours)	<code>rh_run1</code>	0.99 ± 0.02	0.94 ± 0.00	3.41 ± 0.02	1.10 ± 0.05	0.72 ± 0.03
RTAB (RGB) + NoisyGNSS	<code>rh_run2</code>	41.96 ± 1.08	8.33 ± 0.38	1.69 ± 0.07	5.33 ± 0.01	0.41 ± 0.00
RTAB (RGB-D) + NoisyGNSS	<code>rh_run2</code>	59.91 ± 23.48	8.37 ± 0.12	1.93 ± 0.17	5.36 ± 0.08	0.40 ± 0.00
SLPF (ours)	<code>rh_run2</code>	1.00 ± 0.07	0.92 ± 0.05	3.39 ± 0.02	1.13 ± 0.02	0.67 ± 0.03

TABLE II: Operational Row-Level and Headland Transition Metrics for SLPF on Vineyard Traversals `rh_run1` and `rh_run2`. Values are mean $\pm$ standard deviation across seeds 11, 22, and 33. In-row metrics are evaluated exclusively inside crop corridors; transition dynamics are evaluated during headland turns.

Method	Traversal	In-row XT (m)	In-row correct	In-row wrong (s)	Headland XT (m)	Headland recovery (m)
SLPF (ours)	<code>rh_run1</code>	1.10 ± 0.05	0.72 ± 0.03	214.67 ± 19.69	1.63 ± 0.03	2.52 ± 0.21
SLPF (ours)	<code>rh_run2</code>	1.13 ± 0.02	0.67 ± 0.03	230.67 ± 23.63	1.74 ± 0.04	1.71 ± 0.15

TABLE III: Localisation Performance under Simulated GNSS Degradation Stress Profiles. Values are averaged across vineyard traversals `rh_run1` and `rh_run2` over random seeds 11, 22, and 33.

Degradation Profile	Method	Aligned APE (m)	Row Correct	Failure Rate
Nominal	AMCL + NoisyGNSS	1.82	0.58	0.42
Nominal	NoisyGNSS	1.28	0.71	0.29
Nominal	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Nominal	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Nominal	SLPF (ours)	1.00	0.71	0.29
Outage (5 s)	AMCL + NoisyGNSS	1.82	0.58	0.42
Outage (5 s)	NoisyGNSS	1.44	0.71	0.29
Outage (5 s)	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Outage (5 s)	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Outage (5 s)	SLPF (ours)	1.00	0.71	0.29
Outage (10 s)	AMCL + NoisyGNSS	1.82	0.58	0.42
Outage (10 s)	NoisyGNSS	1.44	0.71	0.29
Outage (10 s)	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Outage (10 s)	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Outage (10 s)	SLPF (ours)	0.99	0.71	0.29
Outage (20 s)	AMCL + NoisyGNSS	1.82	0.58	0.42
Outage (20 s)	NoisyGNSS	1.44	0.71	0.29
Outage (20 s)	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Outage (20 s)	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Outage (20 s)	SLPF (ours)	1.01	0.71	0.29
Drift Bias	AMCL + NoisyGNSS	1.82	0.58	0.42
Drift Bias	NoisyGNSS	4.10	0.41	0.59
Drift Bias	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Drift Bias	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Drift Bias	SLPF (ours)	3.68	0.40	0.60
High Gaussian	AMCL + NoisyGNSS	1.82	0.58	0.42
High Gaussian	NoisyGNSS	3.16	0.51	0.49
High Gaussian	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
High Gaussian	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
High Gaussian	SLPF (ours)	1.37	0.65	0.35
Multipath Bursts	AMCL + NoisyGNSS	1.82	0.58	0.42
Multipath Bursts	NoisyGNSS	3.64	0.63	0.37
Multipath Bursts	RTAB RGB + NoisyGNSS	7.24	0.39	0.61
Multipath Bursts	RTAB RGBD + NoisyGNSS	7.68	0.40	0.60
Multipath Bursts	SLPF (ours)	1.20	0.68	0.32

into the new corridor.

During headland transitions, the robot leaves row constraints, resulting in headland cross-track deviations of 1.63 ± 0.03 m and 1.74 ± 0.04 m. As demonstrated in Fig. 5, upon turning into the subsequent row, the filter recovers the correct corridor within a mean recovery distance of 2.52 ± 0.21 m on `rh_run1` and 1.71 ± 0.15 m on `rh_run2`.

C. Robustness to GNSS Degradation and Outages

To assess operational reliability under realistic GNSS failures, we subject the system to seven degradation regimes (Table III):

TABLE IV: Component Ablation Study on Benchmark Traversal `rh_run2` across Seeds 11, 22, and 33. Metric $\Delta APE_{\text{align}}$ denotes difference relative to the full proposed SLPF pipeline.

Configuration	Raw APE (m)	Aligned APE (m)	XT (m)	Row Correct	$\Delta APE_{\text{align}}$ (m)
Full SLPF (proposed)	1.00 ± 0.07	0.92 ± 0.05	1.47 ± 0.03	0.69 ± 0.02	0.00
Non-wall points (point-matching)	1.03 ± 0.10	0.94 ± 0.08	1.33 ± 0.02	0.73 ± 0.02	+0.01
Static GNSS weight	0.98 ± 0.07	0.89 ± 0.04	1.46 ± 0.03	0.69 ± 0.00	-0.03
No pose smoothing	1.00 ± 0.08	0.94 ± 0.06	1.45 ± 0.03	0.69 ± 0.01	+0.01
No background term	1.06 ± 0.03	0.97 ± 0.10	1.42 ± 0.04	0.73 ± 0.02	+0.04
No corridor prior	1.15 ± 0.17	1.06 ± 0.11	1.61 ± 0.08	0.68 ± 0.03	+0.14
No semantic likelihood	1.01 ± 0.06	0.92 ± 0.04	1.34 ± 0.07	0.71 ± 0.04	-0.01
No GNSS prior	4.92 ± 2.03	3.56 ± 1.12	2.84 ± 0.84	0.32 ± 0.13	+2.63

- 1) Nominal: Baseline uncorrupted receiver signals.
- 2) Outages (5 s, 10 s, 20 s): Complete GNSS dropouts where the robot navigates using dead-reckoning and semantic LiDAR alone.
- 3) Drift bias: Continuous cumulative 0.10 m/s lateral drift.
- 4) High Gaussian: Additive zero-mean noise with $\sigma = 5.0$ m.
- 5) Multipath bursts: Sudden 8.0 m discontinuous position jumps lasting 3.0 s.

As detailed in Table III, SLPF exhibits exceptional resilience. Under complete GNSS outages of 5 s, 10 s, and 20 s, SLPF aligned APE remains tightly bounded (1.00 m, 0.99 m, and 1.01 m), maintaining a high row correct fraction of 0.71 and a low failure rate of 0.29. Under high Gaussian noise and multipath bursts, the robust Huber loss successfully attenuates extreme innovations, yielding aligned APEs of 1.37 m and 1.20 m, compared to > 3.16 m for NoisyGNSS and > 7.24 m for RTAB-Map. Only under extreme continuous drift bias (3.68 m APE) does tracking degrade substantially.

D. Component Ablation Study

To isolate the contribution of each algorithmic module, Table IV reports an eight-variant ablation:

- Full SLPF: The proposed complete system (0.92 ± 0.05 m aligned APE, 1.00 ± 0.07 m raw APE).
- Non-wall points: Point-to-point landmark matching without planar wall grouping (0.94 ± 0.08 m aligned APE).
- Static GNSS weight: Standard Gaussian GNSS loss without Huber attenuation (0.89 ± 0.04 m aligned APE).

TABLE V: Localisation Robustness under Perception Occlusions, Landmark Map Decimations, and UAV Map Noise on Traversal `rh.run1` across Seeds 11, 22, and 33.

Evaluation Condition	Aligned APE (m)	Row Correct	Cross-Track (m)	Post-Recovery APE (m)
Baseline (Full Map, 100% Det.)	0.94 ± 0.00	0.71 ± 0.00	1.37 ± 0.01	–
Detection Drop 20%	0.86 ± 0.07	0.73 ± 0.01	1.35 ± 0.03	–
Detection Drop 40%	0.91 ± 0.08	0.71 ± 0.03	1.36 ± 0.01	–
Landmark Removal 30%	0.86 ± 0.08	0.73 ± 0.04	1.38 ± 0.06	0.73 ± 0.02
Landmark Removal 50%	0.93 ± 0.06	0.75 ± 0.01	1.32 ± 0.02	0.80 ± 0.13
UAV Map Noise ($\sigma = 0.25$ m)	1.63 ± 1.05	0.65 ± 0.02	1.62 ± 0.33	–
UAV Map Noise ($\sigma = 0.50$ m)	1.01 ± 0.05	0.68 ± 0.02	1.36 ± 0.09	–

- No pose smoothing: Raw particle weighted mean (+0.01 m error increase).
- No background term: Removing free-space likelihood (+0.04 m error increase).
- No corridor prior: Disabling in-row corridor bounds (+0.14 m error increase, cross-track rises to 1.61 m).
- No semantic likelihood: Relying solely on 2D LiDAR and GNSS (0.92 ± 0.04 m aligned APE).
- No GNSS prior: Complete removal of global GNSS fixes (+2.63 m error increase, row correct collapses to 0.32).

The ablation validates that corridor priors, semantic walls, and GNSS anchoring are essential to achieving robust row disambiguation.

E. Robustness to Perception Occlusion, Decimation, and Map Noise

In practical deployment, camera lenses become soiled with dust, foliage occludes trunks, and prior maps may be derived from coarse aerial surveys. Table V evaluates SLPF under synthetic landmark detection dropouts (20%, 40%), permanent map decimation (30%, 50% removal), and simulated UAV map noise perturbations ($\sigma = 0.25$ m and 0.50 m) reflecting aerial photogrammetry accuracy [23].

Under 40% detection dropouts, aligned APE remains resilient at 0.91 ± 0.08 m with no loss in row correctness (0.71 ± 0.03). When 50% of all map landmarks are removed, SLPF maintains 0.93 ± 0.06 m aligned APE and 0.75 ± 0.01 row correctness, recovering to 0.80 ± 0.13 m post-headland APE. Furthermore, under UAV map perturbations of $\sigma = 0.25$ m and 0.50 m, SLPF maintains stable tracking (1.63 ± 1.05 m and 1.01 ± 0.05 m aligned APE; 0.65 ± 0.02 and 0.68 ± 0.02 row correctness). This confirms that high-precision manual land surveying is not strictly necessary; coarse aerial mapping passes provide sufficient structural guidance for ground robot navigation.

F. Computational Complexity and Runtime Profiling

Real-time efficiency is essential for agricultural autonomous systems. We profiled the complete Python/CUDA SLPF pipeline on an onboard NVIDIA Jetson AGX / RTX workstation.

At a sensor frame stride of 4, the filter processes incoming sensor packets at an average execution time of 76.7 ms per step (≈ 13.0 Hz processed frame rate), supporting an effective sensor input stream of 52.2 Hz. The execution latency is distributed as follows:

- Odometry motion update: 1.2 ms (1.6%).
- LiDAR scan matching: 24.8 ms (32.3%).

- Semantic-wall projection and likelihood: 31.5 ms (41.1%).
- Huber GNSS likelihood evaluation: 3.1 ms (4.0%).
- KLD adaptive resampling and smoothing: 16.1 ms (21.0%).

The particle filter memory footprint remains under 140 MB, confirming that the proposed algorithm is well suited for real-time deployment on standard agricultural mobile computing units.

VI. DISCUSSION AND PRACTICAL INSIGHTS

Our empirical findings offer several concrete guidelines for field robotics in viticulture:

- 1) Decoupled Metric Analysis: Evaluating raw map-frame error alongside aligned APE is essential. Visual SLAM configurations achieved deceptively low aligned RPE (1.20 m) while exhibiting massive raw map errors (> 63 m) due to global corridor aliasing.
- 2) Semantic Walls vs. Unordered Points: Grouping landmarks into continuous planar corridors provides structural rigidity that prevents particles from drifting transversely across trellises.
- 3) Robust GNSS Integration: Single-frequency GNSS cannot reliably identify rows in isolation (≈ 3.0 m error), but when integrated via a Huber loss with threshold $\delta = 3.0$ m, it provides the necessary global anchor while allowing local semantic cues to override transient multipath spikes.

VII. LIMITATIONS

While the presented framework demonstrates strong performance, several operational considerations warrant discussion. First, the method requires an initial topological layout of the vineyard trunks and posts. However, as demonstrated in our robustness evaluations with perturbed maps ($\sigma_{\text{map}} = 0.25\text{--}0.50$ m), centimeter-accurate manual ground surveying is unnecessary; coarse aerial maps extracted from low-altitude drone photogrammetry [23] provide sufficient structural guidance. Second, while robust to partial occlusions, the filter depends on the availability of visible woody structures; in newly planted vineyards with slender vines or during periods of dense, ground-covering canopy overgrowth, landmark detection recall may degrade. Finally, the present evaluation is conducted on a single commercial vineyard; multi-site cross-season validation across diverse trellis architectures (e.g., pergola, Geneva double curtain) remains important future work.

VIII. CONCLUSION

In this paper, we presented the Semantic Landmark Particle Filter (SLPF), a robust localization framework designed to overcome row-level perceptual aliasing and GNSS degradation in vineyards. By combining 2D LiDAR scans, deep semantic detections of stable trunks and support poles structured into row-aligned semantic walls, and a Huber-gated GNSS prior, SLPF achieves sub-metre global accuracy

and reliable corridor identification. Extensive multi-seed experiments across two field traversals demonstrate significant improvements over standard AMCL and visual SLAM baselines, with proven robustness against GNSS outages of up to 20 s, severe sensor noise, and landmark occlusions.

Future work will explore extending the semantic-wall formulation to full 3D multi-camera surround perception, integrating active multi-hypothesis path planning, and performing online semantic mapping during exploratory passes.

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